

10.3 Exploring Absolute Value Equations

Lesson Introduction

 Benchmark	MA.912.AR.4.1 Given a mathematical or real-world context, write and solve one-variable absolute value equations.		 Learning Targets	- I can describe solutions of an absolute value equation from a graph. - I can solve absolute value equations.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			MA.912.AR.4.2 Given a mathematical or real-world context, write and solve one-variable absolute value inequalities. Represent solutions algebraically or graphically.	
 Essential Questions	- How can I determine the number of solutions that an absolute value equation will have? - How can I solve absolute value equations?				
 Lesson Narrative	In this lesson, students solve absolute value equations. Students begin by exploring the graphs of absolute value functions, understanding the graph for situations that produce no solution, one solution, and two solutions. They will then move into solving absolute value equations algebraically and using graphs.				
 Component Summary	10.3.1 Warm-Up (~5 min.)	Critical thinking number puzzle (<i>SE p. 132</i>)	10.3.3 Guided Instruction (10-15 min.)	Introduction to solving an absolute value equation (<i>SE p. 136</i>)	
	10.3.2 Exploration (15-20 min.)	Exploration with the graphs of absolute value functions (<i>SE p. 133</i>)	10.3.4 Guided Practice (15-20 min.)	Practice solving absolute value functions (<i>SE p. 137</i>)	
	10.3.5 Wrap-Up (5 min.)	Note to a friend activity (<i>SE p. 137</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Chart paper for an anchor chart		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may solve an existing equation, and then just change the value of the original answer to create the second answer. For example, in 10.3.3 Guided Instruction, students see 5 and -5 as solutions for question 1. Students may then see 5 as the solution for question 2 and assume the “other” answer is also -5.	10.3.2 Exploration is designed to involve student discovery. Be careful not to give away too much information formally during this process. Provide opportunity for students to make mistakes, explore, ask questions, and engage with peers during this component.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share During 10.3.2 Exploration, students will work with a partner to explore the graphs of absolute value functions. Leverage this routine to allow students an opportunity to try on their own and then compare their work and process with a partner.	MA.K12.MTR.4.1: Discussion and Reflection 10.3.2 Exploration provides students with a collaborative exploration with the graphs of absolute value functions. During this activity, students will construct possible arguments based on evidence and justify results by explaining methods and processes.
 Differentiate Instruction	During 10.3.2 Exploration, consider differentiating question 2 by providing a visual entry-point for learning. In question 2, students are exposed to two functions with different colors on the same coordinate plane. Consider re-creating with two coordinate planes, each with one of the graphs.	
Lesson Notes:		

10.3.1 Warm-Up: Critical Thinking

Component Overview: Students will complete a critical thinking activity where they will create a true absolute value equation using digits with limitations.



10.3.1 Warm-Up: Critical Thinking

- Place a digit in each blue box to make the statement true. Use only digits 1 to 9, at most one time each.

$$\boxed{} + \boxed{} = |\boxed{} - \boxed{}|$$

Answers will vary. $3 + 4 = |2 - 9|$. $1 + 2 = |8 - 5|$.



Consider leveraging a Think-Pair-Share routine with this activity. Give students a few minutes to try this on their own and then compare their work with a partner. Have multiple students share their approaches and strategies in a whole group setting so students can hear different methods.

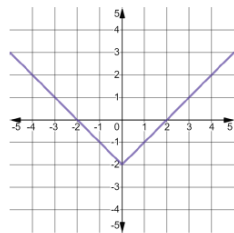
10.3.2 Exploration: Graphs of Absolute Value Functions

Component Overview: Students will complete a five-question exploration with the graphs of absolute value functions. For this activity, students will need a partner.

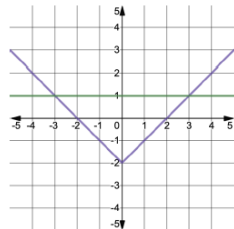


10.3.2 Exploration: Graphs of Absolute Value Functions

- The graph of the absolute value equation $y = |x| - 2$ is shown.



The graph of $y = |x| - 2$ can be used to find the solution to $1 = |x| - 2$.



- What is the equation of the green line?
 $y = 1$
- What are the points of intersection for the two equations?
 $(-3, 1)$ and $(3, 1)$
- Substitute the x -coordinates of the points of intersection into the equation $1 = |x| - 2$.

$1 = x - 2$ $1 = -3 - 2$ $1 = 3 - 2$ $1 = 1$	$1 = x - 2$ $1 = 3 - 2$ $1 = 3 - 2$ $1 = 1$
---	--



What do you notice about the absolute value graph? What do you wonder?



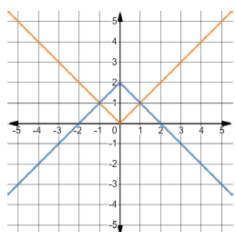
What do the points of intersection represent?

10.3.2 Exploration: Graphs of Absolute Value Functions (continued)

- d. Discuss with your partner what you noticed when the x -coordinates were substituted into the equation. Write a summary of your discussion.

Answers will vary. I noticed that once the absolute value of the number is determined, the remainder of the steps and answer are the same.

2. The graphs of $y = |x|$ and $y = -|x| + 2$ are shown on the coordinate grid.



- a. Decide who will be Partner A and who will be Partner B, then complete the table.

Partner A		Partner B	
What is/are the value(s) of y when $x = 2$ on the blue graph?	0	What is/are the value(s) of y when $x = 0$ on the orange graph?	0
What is/are the value(s) of x when $y = 1$ on the blue graph?	-1 and 1	What is/are the value(s) of x when $y = 0$ on the orange graph?	0
What is/are the value(s) of x when $y = 3$ on the blue graph?	N/A	What is/are the value(s) of x when $y = -3$ on the orange graph?	N/A

- b. What was true when there was only one value of x for the given y -value?

It was the vertex.



Consider differentiating question 2 by providing a visual entry-point for learning. In question 2, students are exposed to two functions with different colors on the same coordinate plane. Consider recreating with two coordinate planes, each with one of the graphs.



- What is the relationship between your answer in question 2b and the axis of symmetry?
- Why is there no value of x when $y = 3$ and $y = -3$?

10.3.2 Exploration: Graphs of Absolute Value Functions (continued)

- c. What was true when there were two values of x for the given y -value?

There were two solutions to the absolute value function that are equidistant from the vertex.

- d. What was true when there were no values of x for the given y -value?

There were no solutions to that absolute value function.

- e. Discuss with your partner the circumstances under which an absolute value function would have each type of solution.

Answers will vary.

3. Complete each sentence.

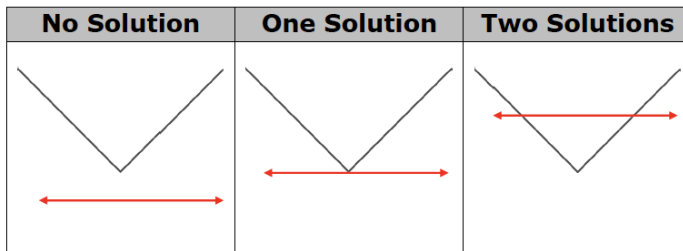
No Solution
There will be no solution when the y -value ... <i>is less than the y-value of the vertex (or greater than when the absolute value graph is negative).</i>
One Solution
There will be one solution when the y -value ... <i>is the y-value of the vertex.</i>
Two Solutions
There will be two solutions when the y -value ... <i>is greater than the y-value of the vertex (or less than when the absolute value graph is negative).</i>



Consider taking the information from question 3 and creating an anchor chart or note-taking reference guide for students. Creating this centralized reference for students can support their learning of this content throughout the remainder of the unit.

10.3.2 Exploration: Graphs of Absolute Value Functions (continued)

4. Sketch a graph for each solution type.



Answers will vary for no solution and two solutions. Examples shown. Sketches should meet the accurate conditions for each solution type.

5. Four absolute value equations are shown.

- a. Work with your partner to determine the x -values that make the equations true.

$x = 2 $	$x = -2 $
2	2
$ x = 3$	$ x = -3$
-3 or 3	N/A

- b. What did you notice about the four equations?

Answers will vary.



If you created a reference guide or anchor chart for question 3, add the additional information from question 4 to that document. Consider allowing various sketches that are similar, but are not identical, so that students can see the common features/conditions.



At the conclusion of this activity, spend some extra time in a discussion for question 5. Each of these can represent a misconception that students have. Engage students in thinking through what each of them represents and what they mean.



Why is it not possible for x to equal -2 ?

For $|x| = 3$, why does x equal -3 and 3 ?

Why is it not possible for $|-3| = -3$?

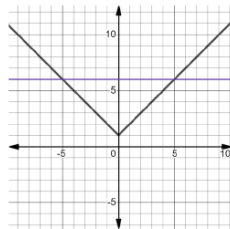
10.3.3 Guided Instruction: Solving an Absolute Value Equation

Component Overview: Students will learn how to solve absolute value equations using two unique questions with multiple parts.



10.3.3 Guided Instruction: Solving an Absolute Value Equation

1. The graphs of $y = |x| + 1$ and $y = 6$ are shown on the coordinate grid.



- a. Use the graph to determine the solution to $|x| + 1 = 6$.

$5; -5$

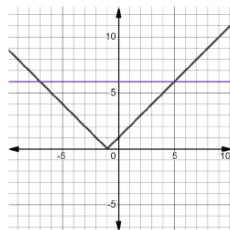
- b. Solve $|x| + 1 = 6$ algebraically.

$$|x| + 1 = 6$$

$$|x| = 5$$

$$x = 5; -5$$

2. The graphs of $y = |x + 1|$ and $y = 6$ are shown on the coordinate grid.



- a. Use the graph to determine the solution to $|x + 1| = 6$.

$x = -7; 5$

- b. Solve $|x + 1| = 6$ algebraically.

$$|x + 1| = 6$$

$$x + 1 = 6$$

$$x = 5$$

$$x + 1 = -6$$

$$x = -7$$



For this activity, consider guiding students through question 1a as the graphing component, and then question 1b as the algebraic part. For question 2, provide the narrative information and release students to try question 2a on their own. Bring them back together to ensure they understand that method. Then, repeat this process for question 2b algebraically.



What are some of the key differences between using a graph to determine the solution vs. algebraically?

Which method did you prefer?

10.3.4 Guided Practice: Solving Absolute Value Functions

Component Overview: Students will complete four practice questions on solving absolute value functions algebraically.



10.3.4 Guided Practice: Solving Absolute Value Functions

Solve each absolute value equation algebraically.

1. $|w| - 42 = -37$
5; -5

2. $|x - 12| = 23$
35; -11

3. $16 = |y| + 5$
11; -11

4. $|9z| = 15$
 $\frac{5}{3}; -\frac{5}{3}$



Consider using this time to work with small groups of students who struggled with questions 1b and 2b from 10.3.3 Guided Instruction.



What are the similarities and differences in solving an absolute value equation compared to a linear equation?

10.3.5 Wrap-Up: Cool Down

Component Overview: Students will write a note to an absent student explaining what they missed from this lesson.



10.3.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to them explaining how to solve absolute value equations.

Answers will vary.



Encourage students to be creative in their application or representation of their responses.

- Students can use social media to record mini lessons for teaching others (and make money doing it).
- Students can record their responses verbally or find a partner to “interview” about the lesson concepts. Determining what questions need to be asked often exhibits more conceptual understanding than answering them.
- Students can act as the teacher for a younger (or older) sibling and report their results.